

Review

Biochar-Based Materials for Catalytic CO₂ Valorization

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Abstract: Biochar-based materials have gathered increasing attention as sustainable catalysts for carbon dioxide (CO₂) valorization, offering a green alternative to traditional metal-based systems. Produced from renewable biomass through pyrolysis, biochar possesses key features—such as high surface area, rich porosity and tunable surface chemistry—that make it particularly suited for heterogeneous catalysis. This review highlights recent advances in the use of biochar-derived catalysts for key CO₂ conversion reactions, focusing on cycloaddition to epoxides, dry reforming of methane and catalytic biomass upgrading. Emphasis is given to the role of biochar's origin and preparation methods, which critically influence its structure, surface functionality and catalytic performance. Feedstocks rich in mineral content or oxygenated groups, for instance, can enhance CO₂ activation and product selectivity. Furthermore, tailored modifications—such as doping with heteroatoms or supporting metal nanoparticles—further boost catalytic activity and stability by tuning acid–base behavior, while maintaining low toxicity and cost-effectiveness. Compared to conventional catalysts, biochar-based systems offer advantages in low cost, recyclability and resistance to deactivation. Challenges remain in standardizing production methods, controlling structural variability, minimizing metal leaching and scaling up. This review presents biochar as a versatile, renewable platform for CO₂ utilization, highlighting the importance of rational design, feedstock selection and functionalization strategies for developing efficient, sustainable catalytic systems, in line with green chemistry and circular economy principles.

Keywords: biochar; catalysis; carbon dioxide; epoxide cycloaddition; dry reforming of methane; biomass upgrade



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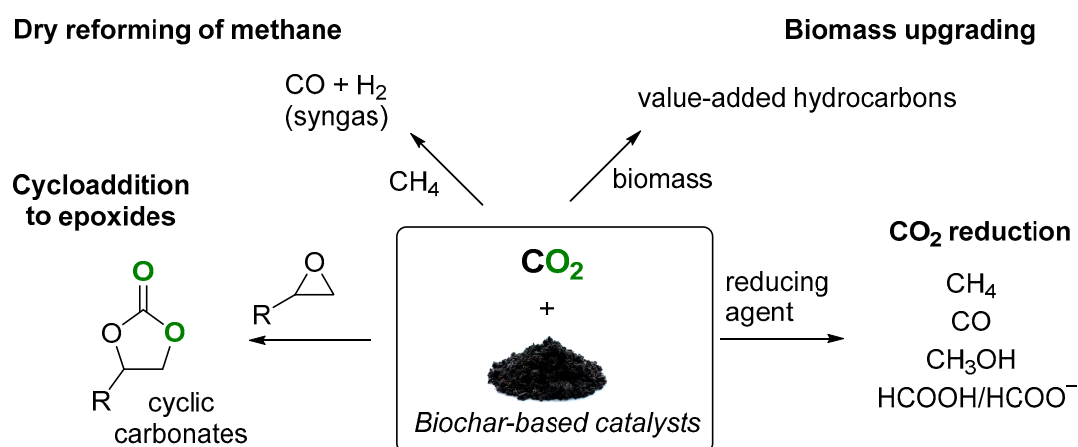
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1. Introduction

Biochar is a carbon-rich material, produced through the pyrolysis of biomass, which has garnered significant attention due to its diverse applications in environmental and energy-related fields [1–3]. It has been widely used in soil amendment/fertilization [4], carbon sequestration [5–7], pollution remediation [8–10] and as an advanced material for supercapacitors and batteries [11–14]. In addition, biochar has emerged as a promising catalyst and catalyst support in various catalytic transformations, owing to its sustainable origin, well-defined structural architecture and highly tunable surface chemistry [15–20]. The catalytic efficiency of biochar arises from a combination of intrinsic characteristics

that make it an attractive alternative to conventional catalysts. These include: (a) high porosity and large surface area, which facilitate the dispersion of active catalytic sites and enhance reactant adsorption; (b) surface functionalization potential, with abundant oxygen-containing groups (e.g., hydroxyl, carboxyl) that can be modified or functionalized to introduce specific catalytic functionalities; (c) exceptional thermal and chemical stability, allowing biochar to withstand a wide range of reaction conditions while maintaining its structural integrity, making it a robust catalytic material [21]. These materials often function as support for metal nanoparticles or as carriers of active functional groups, which significantly enhance their catalytic activity and selectivity [22–24]. Biochar-based catalysts have been successfully employed in numerous synthetic transformations, including C–C coupling reactions [25], oxidation [26,27], reduction [28] and biomass valorization [29] processes. In recent years, growing efforts have been directed toward utilizing biochar in the catalytic conversion of carbon dioxide (CO_2) into value-added chemicals and fuels [30–32]. This approach aligns with the principles of green chemistry [33–37] and circular economy [38], offering a sustainable pathway to mitigate CO_2 emissions while producing high-value products. Given its renewable nature, cost-effectiveness and tunable properties, biochar presents a compelling alternative to organometallic-based homogeneous catalysts and to conventional heterogeneous catalysts, particularly in processes where sustainability and economic feasibility are critical factors.

Key reactions involving CO_2 transformation promoted by biochar-derived catalysts are summarized in Scheme 1. They include (1) cycloaddition of CO_2 to epoxides to produce cyclic carbonates, which find numerous applications as green solvents or as valuable intermediates in polymer synthesis; (2) dry reforming of methane (DRM) to generate synthesis gas (syngas), a crucial feedstock for the chemical industry; (3) biomass catalytic upgrading reactions, which generate high-value products and fuels; (4) CO_2 reduction reactions, including catalytic hydrogenation, photocatalytic reduction and electrocatalytic reduction, to produce methane, carbon monoxide, methanol or formate, which allow for the production of synthetic natural gas, syngas, for Fischer-Tropsch processes and fuels, contributing to the development of carbon-neutral energy cycles [39,40].



Scheme 1. Examples of CO_2 valorization reactions using biochar-based catalysts.

This review offers a focused and critical overview of recent advancements in biochar-based catalysts for CO_2 valorization reactions. We specifically explore their application in three key areas: catalytic CO_2 cycloaddition to epoxides, dry reforming of methane, and biomass catalytic upgrading. The review is organized by the synthesis methods of these biochar-based catalysts within each reaction type, highlighting their achievements, potential and current challenges, along with future prospects.

2. Cycloaddition of CO₂ to Epoxides

The cycloaddition of CO₂ to epoxides is a particularly attractive transformation, as it produces cyclic carbonates with 100% atom economy [41–45]. Cyclic carbonates are considered valuable intermediates in polymer synthesis, and they find numerous applications, namely as electrolytes for lithium-ion batteries, and as bioactive compounds [46]. Conventionally, this reaction requires homogeneous catalysts [47–49], such as metal complexes, which often suffer from issues related to recyclability, cost and negative environmental impact. To overcome these challenges, heterogeneous catalysts, namely, those based on renewable biochar, have been recently used to promote such reactions in a more sustainable manner. Table 1 summarizes the studies, reported in the literature, that describe the utilization of biochar-based catalysts in CO₂ cycloaddition reactions to epoxides.

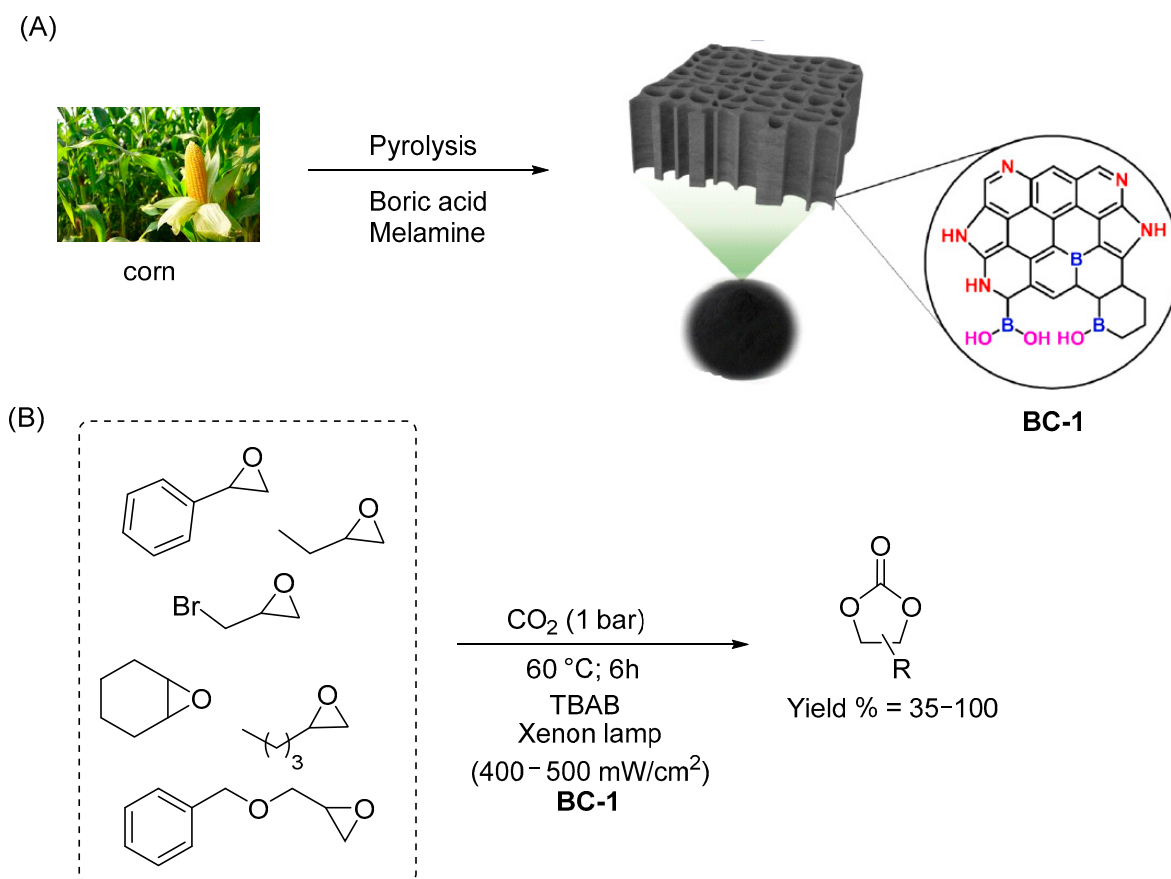
Table 1. CO₂ cycloaddition to epoxides catalyzed by biochar-based materials.

Entry	CAT	BET Area (S _{BET}) (m ² /g)	Co-CAT	T (°C)	P (bar)	t (h)	Reuse Cycles	Yield (%)	Ref.
1	Biochar prepared from corn stalk (BC-1)	166	TBAB	60	1	6	10	up to 100 ^a	[50]
2	Biochar prepared from chitosan (BC-2)	259	CTAB; NH ₄ Cl; NaCl	120–140	1	4–6	5	80–97 ^b	[51]
3	Biochar prepared from soft and hardwood (BC-3 and BC-4)	231	TBAB; PPNCl; PPNN ₃	110	10–20	4–16	5	22–100 ^c	[52]
	Biochar prepared from chitosan/cellulose (BC-5)	660	none	60–120	10–30	12	5	0–99 ^d	[53]
5	Biochar prepared from natural polysaccharides (BC-6)	442	none	70	3	7	5	0–96 ^e	[54]
6	Biochar prepared from chitosan (BC-7)	635	none	150	15	15	3	80–99 ^f	[55]
7	Biochar prepared from coffee grounds (BC-8)	654	none	150	20	12	5	95 ^g	[56]
8	Biochar prepared from eucalyptus waste (BC-9)	527	none	80–120	20	24	4	22–58 ^h	[57]

^a best results for 1,2-epoxybutane, epibromohydrin, styrene oxide, phenyl glycidyl; ^b best result for epichlorohydrin; ^c best result for glycidol; ^d best result for epibromohydrin; ^e best result for epichlorohydrin; ^f best result for epichlorohydrin and glycidol; ^g for epichlorohydrin; ^h for epichlorohydrin.

Ding and Yao [50] developed a biochar-based catalyst (**BC-1**), using corn stalk powder, boric acid, and melamine through a pyrolysis process (Scheme 2A). The study emphasizes the fine-tuning of biochar active sites via one-step and two-step calcination, while preserving the tubular structure of the corn stalk. Notably, **BC-1**, synthesized through a one-step doping process, exhibited significantly enhanced photocatalytic activity under Xe lamp irradiation (400–500 mW/cm²) using TBAB as a co-catalyst at 1 bar CO₂ pressure (Scheme 2B). This system achieved an impressive cyclic carbonate yield of up to 100% using a range of epoxides (Table 1, entry 1).

The biochar-based catalyst **BC-1** exhibited exceptional recyclability, maintaining its catalytic efficiency under light irradiation over ten consecutive catalytic cycles (Figure 1). Contrast experiments and photocurrent tests revealed that the high activity of **BC-1** arises from the synergistic effect between photocatalytic and photothermal properties. This study offers valuable insights into the design and development of biochar-based photothermal catalysts for the efficient conversion of CO₂ into high-value products.



Scheme 2. (A) Pyrolytic synthesis of biochar-based catalyst **BC-1** from corn stalk powder, boric acid, and melamine; (B) CO₂ cycloaddition to epoxides using **BC-1** as catalyst. Adapted from reference [50] with permission from Elsevier, Copyright 2025.

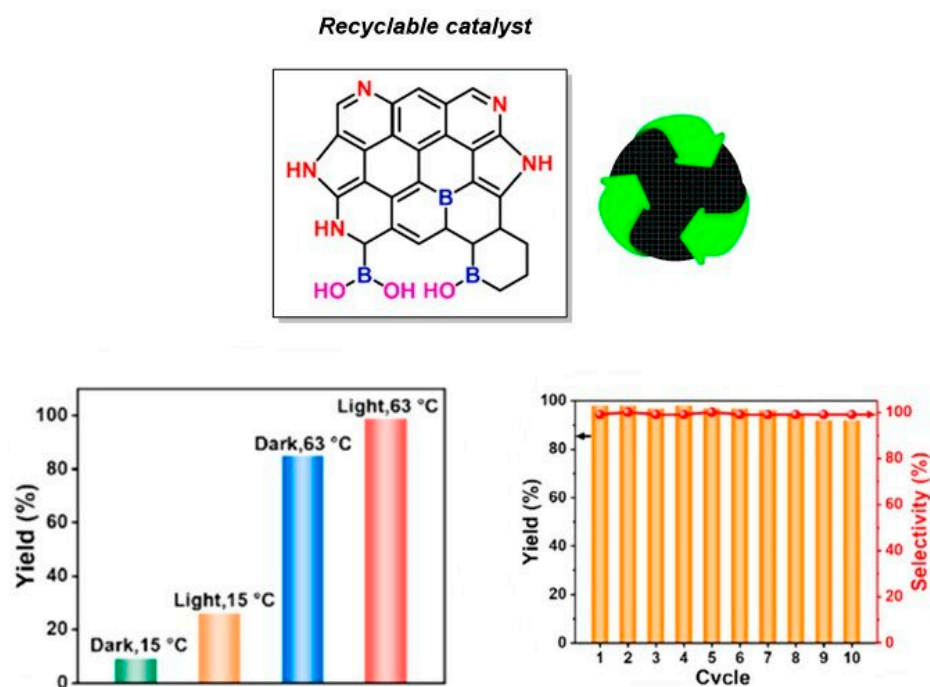
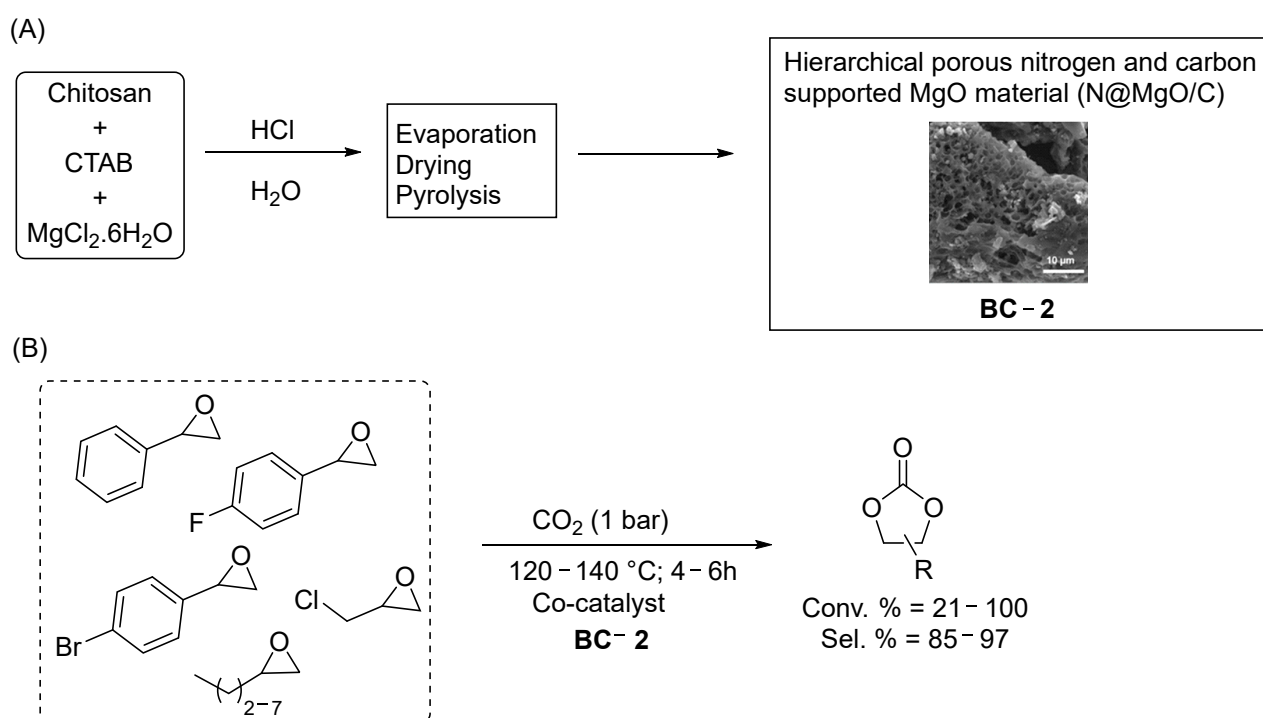


Figure 1. Recycling performance of **BC-1** in CO₂ cycloaddition to epoxides, under light irradiation, demonstrating high efficiency over ten cycles. Adapted from reference [50] with permission from Elsevier, Copyright 2025.

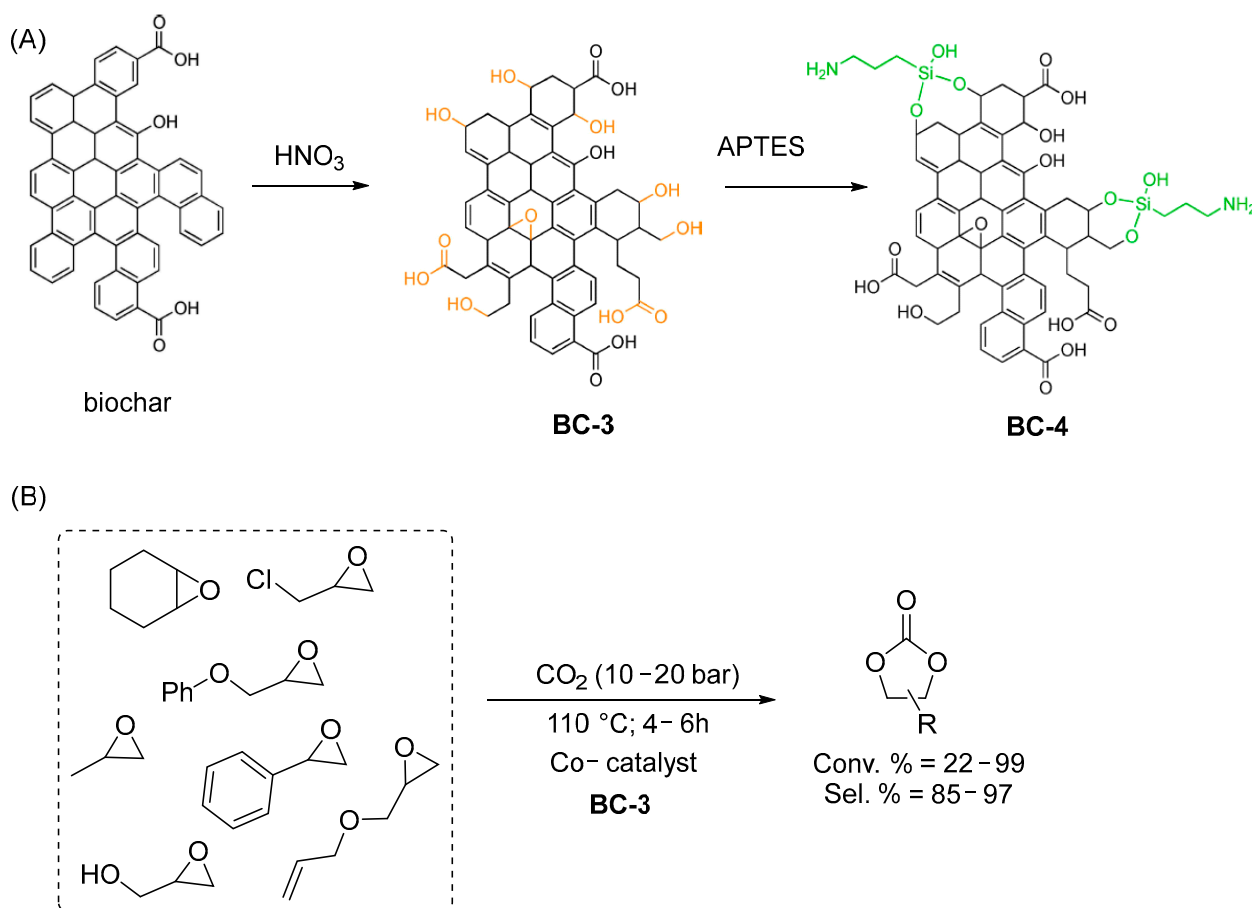
Grafouté and Biradar [51] reported a straightforward approach for synthesizing a novel hierarchical porous N/C-supported MgO catalyst (**BC-2**) using abundant bio-waste and magnesium chloride. To facilitate catalyst formation, cetyltrimethylammonium bromide (CTAB) and $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ were introduced into an aqueous HCl solution of chitosan, leading to the formation of a homogeneous Mg^{2+} -containing complex. The reaction mixture was then heated to 100 °C until water evaporation was complete, followed by drying at 80 °C. Finally, this material was submitted to a pyrolysis process (Scheme 3A). The material was characterized by XRD, Raman and XPS analyses, as well as FESEM and N_2 adsorption-desorption, revealing a high surface area ($258.93 \text{ m}^2/\text{g}$), a pore volume of $0.063 \text{ cm}^3/\text{g}$, and a narrow pore diameter (1.42 nm). Its catalytic performance was assessed in the solvent-free cycloaddition of CO_2 to several epoxides (Table 1, entry 2) at just 1 bar pressure and a temperature of 120 °C–140 °C (Scheme 3B). The catalyst **BC-2** exhibited high stability and reusability, keeping its structural integrity and catalytic efficiency for up to five cycles without morphological changes [51].



Scheme 3. (A) General synthesis of N/C-supported MgO Material (**BC-2**) from chitosan and magnesium chloride; (B) CO_2 cycloaddition to epoxides catalyzed by porous N-doped carbon-supported MgO. Adapted from reference [51] with permission from Elsevier, Copyright 2022.

MacQuarrie and Kerton [52] utilized residues from the forestry, pulp and paper industries—specifically birch and pine—to produce biochar, which was subsequently oxidized with HNO_3 to yield carboxylic acid-functionalized biochar **BC-3**. This was then treated with 3-aminopropyltriethoxysilane (APTES) to afford **BC-4** (Scheme 4A). Characterization by FT-IR, TGA and elemental analysis confirmed its structural properties, while catalytic tests demonstrated good activity in the cycloaddition of CO_2 with various epoxides under mild conditions (Table 1, entry 3). Despite the significantly higher surface area of hardwood-derived biochar, no substantial differences in catalytic performance were observed between the two oxidized biochar materials. Both materials **BC-3** and **BC-4** were applied as reusable catalysts for CO_2 cycloaddition reactions to epoxides (Scheme 4B). The authors concluded that functionalization with 3-aminopropyltriethoxysilane (APTES) was shown to deactivate the catalyst, demonstrating the essential role of surface acid groups

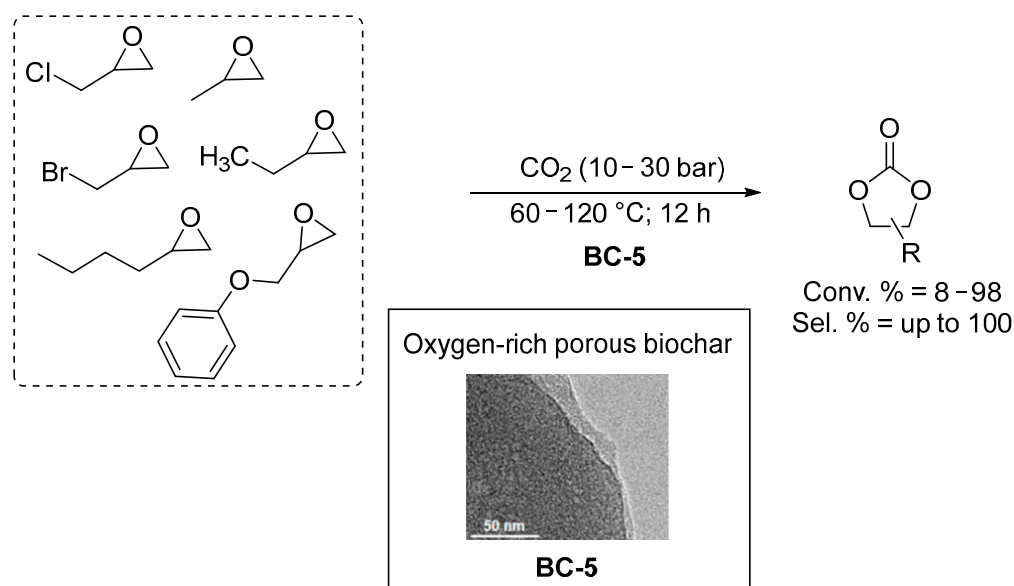
to catalyze the cycloaddition reaction, by facilitating substrate activation via hydrogen bonding. Furthermore, the catalyst **BC-3** demonstrated excellent recyclability and stability over five catalytic cycles. This study shows the potential of oxidized biochar as an effective and reusable catalyst for CO₂ cycloaddition reactions to epoxides, enhancing its catalytic potential, proving that modified biochar from both hardwood and softwood can serve as an efficient catalyst for application of CO₂ in carbonate synthesis.



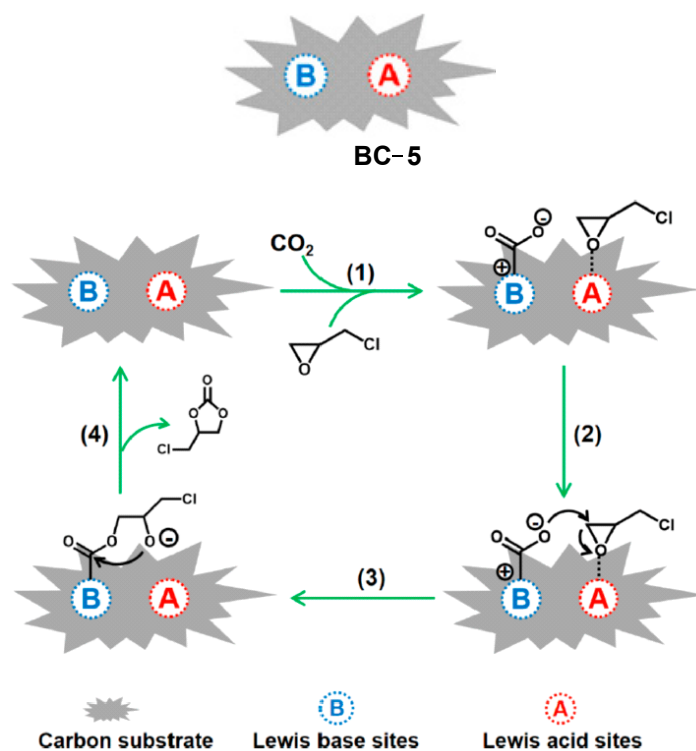
Scheme 4. (A) Synthesis of oxidized biochar (**BC-3**) and APTES-functionalized biochar (**BC-4**); (B) cyclic carbonate synthesis from CO₂ and epoxides using **BC-3** as catalyst.

Jiang's group [53] developed a nitrogen- and oxygen-rich porous biochar (**BC-5**) via a one-pot pyrolysis of chitosan. The material was characterized using EDS, N₂ adsorption-desorption, BET, elemental analysis and XPS. Given the growing need for efficient catalysts in CO₂ fixation, **BC 5** was evaluated for the cycloaddition of CO₂ to epoxides without requiring a co-catalyst or solvent (Table 1, entry 4). The authors described a synergistic effect between nitrogen and oxygen dopants, which enhanced the catalytic performance of the material in CO₂ cycloaddition reactions (Scheme 5).

According to the study, the mechanistic pathway for the cycloaddition of CO₂ to epichlorohydrin involves four key steps (Scheme 6). First, CO₂ and the epoxide are adsorbed and activated on the catalyst surface (step 1). Next, an oxyanion intermediate is formed (step 2), facilitating the subsequent synthesis of the cyclic carbonate (step 3). Finally, the cyclic carbonate desorbs from the catalyst surface (step 4), completing the reaction cycle. Additionally, DFT calculations demonstrated that the exceptional catalytic efficiency of the N,O-co-doped porous biochar arises from its strong CO₂ and epoxide adsorption capabilities. This is driven by its high specific surface area and the activation of CO₂ facilitated by its well-integrated acid-base active sites [53].



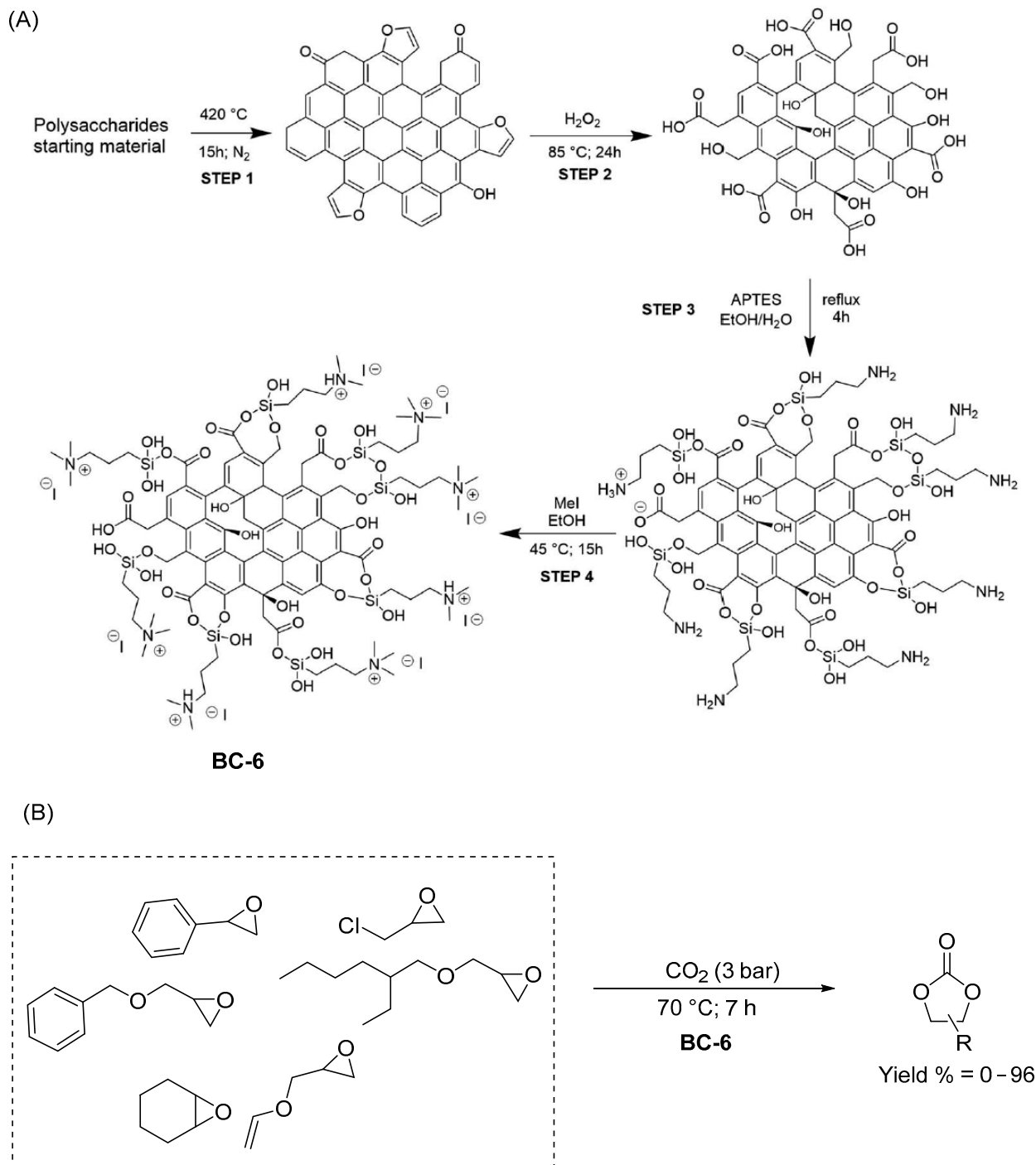
Scheme 5. Catalytic CO₂ cycloaddition with epoxides for cyclic carbonate synthesis using **BC-5** as catalyst. Adapted from reference [53] with permission from Elsevier, Copyright 2025.



Scheme 6. Mechanistic pathway for the cycloaddition of CO₂ using biochar-based bifunctional catalyst **BC-5**. (1) CO₂ and epoxide adsorption; (2) oxyanion formation; (3) cyclic carbonate synthesis; (4) cyclic carbonate desorption. Adapted from reference [53] with permission from Elsevier, Copyright 2022.

Parodi and Vagnoni [54] synthesized biochar materials from three polysaccharide sources—cellulose, cellulose acetate and starch—which were then chemically modified and utilized as bifunctional heterogeneous catalysts for CO₂ cycloaddition with epoxides (Scheme 7). To optimize the biochar-based catalyst synthesis, the authors conducted a systematic study on pristine cellulose, monitoring the process through elemental analysis (Scheme 7A). In this study, higher temperatures led to the formation of more aromatic structures, while shorter reaction times or lower temperatures produced fragile materials

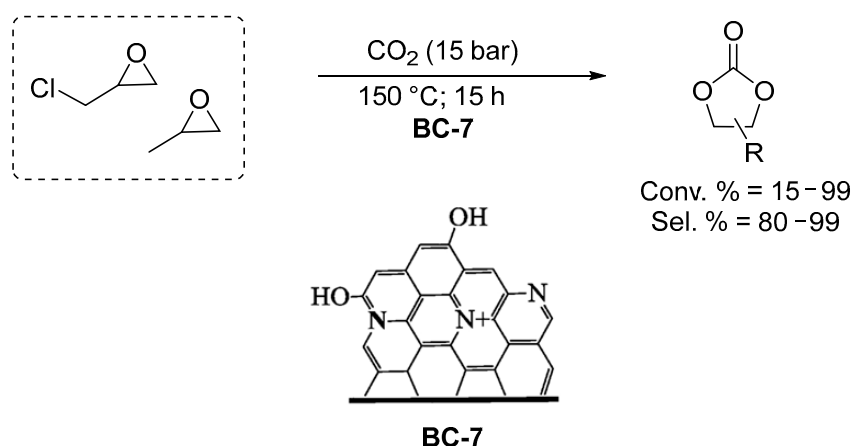
incapable of maintaining their 3D framework. After a series of optimization experiments, the resulting catalyst **BC-6** was characterized using XPS, SEM, Raman and FT-IR analyses. It was then tested across a range of epoxides under mild reaction conditions (Table 1, entry 5), achieving high yields of up to 96% of cyclic carbonates when terminal epoxides were used as substrates (Scheme 7B). Additionally, **BC-6** exhibited excellent recyclability, retaining its catalytic performance for up to five consecutive cycles.



Scheme 7. (A) Preparation of bifunctional heterogeneous catalyst **BC-6** from biomass and waste polysaccharides; (B) CO_2 cycloaddition to epoxides catalyzed by **BC-6**.

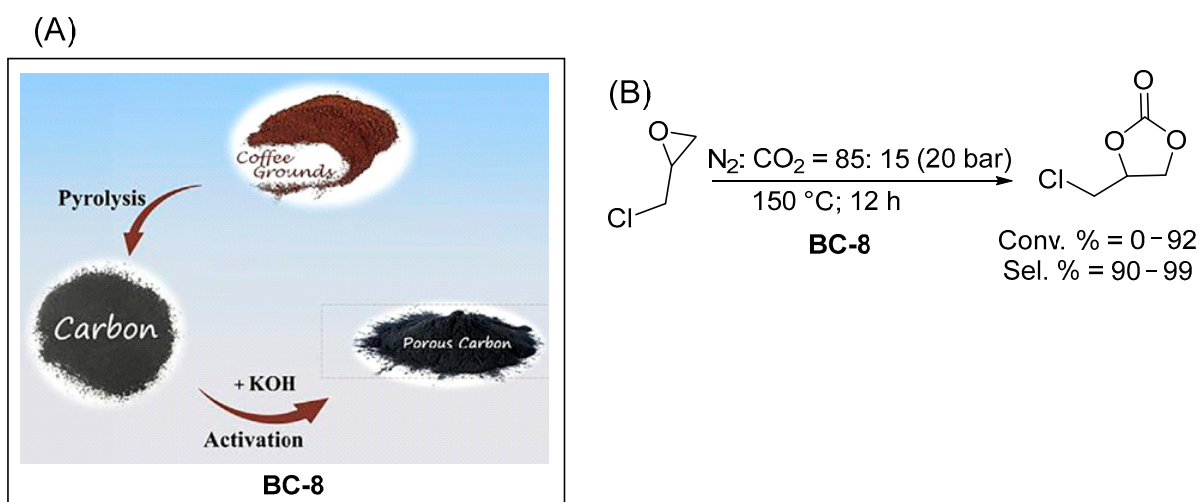
Different nitrogen-doped biochar-based catalysts (**BC-7**) were developed by Konwar [55], through a simple one-step phosphoric acid activation of nitrogen-containing

biomass precursors (chitosan, *Jatropha curcas* and *Mesua ferrea* DOWCs). These materials were characterized using N₂ physisorption, chemisorption, XPS, SEM, TEM, XRD, FT-IR and Micro-Raman spectroscopy. The biochar material **BC-7** was applied as a catalyst in the cycloaddition of CO₂ to epoxides (Scheme 8), producing cyclic carbonates with yields of up to 99%, without the need of a solvent or co-catalyst (Table 1, entry 6). Mechanistic studies revealed that the high catalytic efficiency of **BC-7** stems from its Lewis basic sites, specifically pyridinic, pyridonic and quaternary nitrogen species, which as observed by other authors play a crucial role in CO₂ activation.



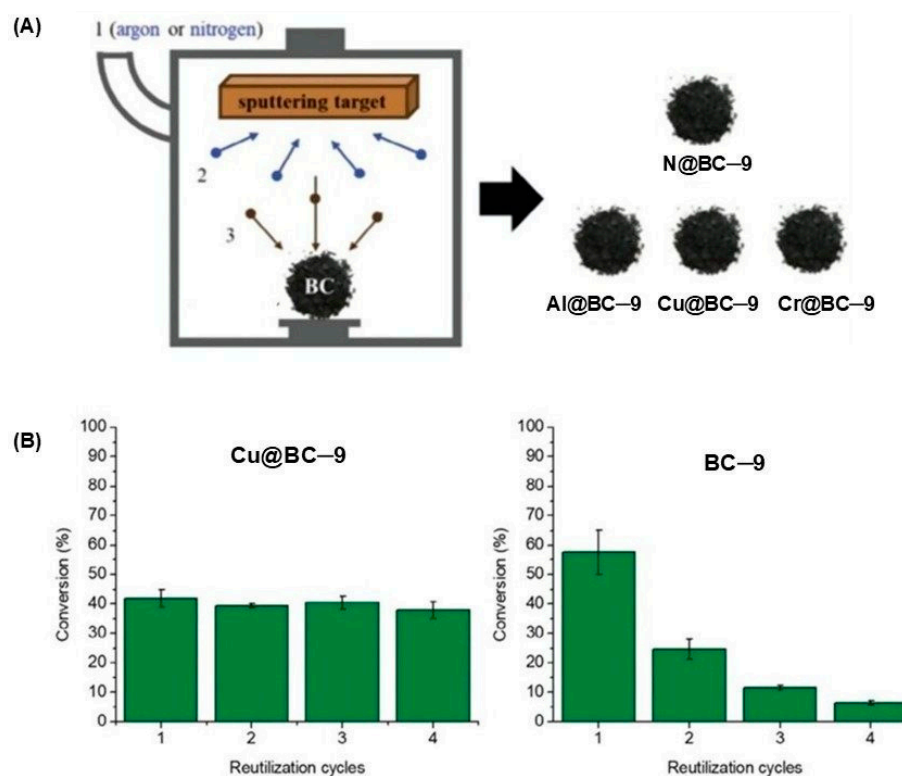
Scheme 8. Synthesis of cyclic carbonates from epoxides and CO₂, using renewable N-doped biomass-derived carbon-based catalyst **BC-7**.

N-enriched microporous active carbons (**BC-8**) were synthesized by Wang [56], using coffee grounds as a biomass precursor and KOH as an activator (Scheme 9A). The resulting materials were characterized by N₂ adsorption–desorption, XRD, Raman, SEM and XPS and evaluated as catalysts for the cycloaddition of CO₂ with epichlorohydrin as a model epoxide (Scheme 9B). The biochar-based catalyst **BC-8** exhibited a remarkable efficiency to yield cyclic carbonates (Table 1, entry 7) and excellent recyclability, maintaining both selectivity and catalytic activity over five consecutive cycles.



Scheme 9. (A) Synthesis of N-enriched microporous activated carbons derived from coffee grounds (**BC-8**); (B) CO₂ cycloaddition to epichlorohydrin catalyzed by **BC-8**. Adapted from reference [56] with permission from Elsevier, Copyright 2020.

Carrilho and Ferreira [57] developed a series of metal- and nitrogen-coated biochar-based catalysts from *Eucalyptus globulus* wood through a novel magnetron sputtering approach (Scheme 10A). The biochar (BC-9), derived from the forest residue, was functionalized with nitrogen and transition metals (Al, Cu, Cr), generating bifunctional catalysts (Al@BC-9, Cu@BC-9, Cr@BC-9) which preserved the high porosity of the initial material (up to 534 m²/g).



Scheme 10. (A) Synthesis of N- and metal-coated biochar materials by magnetron sputtering technique; (B) Reutilization cycles of eucalyptus-derived biochar catalysts BC-9 and Cu@BC-9 in CO₂ cycloaddition to epichlorohydrin. Adapted from reference [57] with permission from John Wiley & Sons Inc., Copyright 2025.

These catalysts were applied in the cycloaddition of CO₂ to epichlorohydrin under mild conditions (20 bar CO₂, 120 °C), without any solvent or co-catalyst. Among them, the unmodified biochar (BC-9) achieved the highest conversion (58%) (Table 1, entry 8), but exhibited progressive deactivation upon reuse [57]. In contrast, Cu@BC-9 maintained stable catalytic performance over four consecutive cycles, with a consistent 42% conversion and >99% selectivity (Scheme 10B), demonstrating improved recyclability due to surface coating with the metal. Although the initial activity of metal-coated samples was slightly lower, their enhanced stability under reaction conditions underscores the benefits of surface metal deposition in prolonging catalyst lifetime. Elemental and surface analyses (ICP and SEM-EDS) confirmed that metal deposition occurred mainly on the outer surfaces of the biochar particles. This localized modification was sufficient to impact the catalyst durability. This work highlights the potential of magnetron sputtering as a scalable method for the synthesis of stable, reusable biochar-based catalysts for CO₂ upgrading reactions. It also emphasizes the relevance of the biomass source and surface composition, particularly metal incorporation, in influencing efficiency and selectivity in biochar-based catalysts [57].

Overall, BC-1, despite its lower surface area ($S_{\text{BET}} = 166 \text{ m}^2/\text{g}$), appears exceptionally promising due to its 100% yield and high recyclability under relatively mild conditions, benefiting from photocatalytic activation. BC-5 and BC-7, which showed S_{BET} areas of 660

and 654 m²/g, respectively, are noteworthy for achieving high yields without co-catalysts, highlighting the effectiveness of intrinsic N and O doping for synergistic acid–base site creation and CO₂ activation. The work on BC-9 ($S_{\text{BET}} = 527 \text{ m}^2/\text{g}$) with metal coating provides crucial insights into improving catalyst durability, a common challenge in catalysis. In sum, modulating the electronic structure, acid–base properties or CO₂ adsorption energy through heteroatom doping or metal additions (e.g., Ni, Co, Mg) fundamentally dictates the efficiency of CO₂ cycloaddition to epoxides. These strategic compositional changes allow the catalyst to precisely tune its Lewis acidity to activate the epoxide ring and its basicity to activate CO₂ (Scheme 6), optimizing molecular interactions and lowering reaction barriers. By fine-tuning CO₂ adsorption energy, the catalyst ensures the gas is sufficiently activated for reaction without being too strongly bound, which would hinder product desorption and catalyst turnover. It could also be withdrawn that surface areas do not play a crucial role in the activity of the biochar catalysts, as results were largely independent from the surface area. Essentially, these modifications provide comprehensive control over how the catalyst binds and activates both reactants, directly influencing the reaction’s efficiency, selectivity and overall catalytic performance [53].

3. Dry Reforming of Methane

Rising atmospheric concentrations of carbon dioxide (CO₂) and methane (CH₄) underscore the urgent need for technologies that can effectively reduce greenhouse gas emissions. One promising strategy involves converting these gases into valuable fuels and chemicals via syngas (a mixture of CO and H₂) production. This can be achieved through a range of thermochemical and electrochemical pathways, including partial oxidation, steam reforming, autothermal reforming, oxidative coupling and dry reforming of methane (DRM). Among these, DRM stands out for its dual benefit: it simultaneously consumes both CO₂ and CH₄—two of the most potent greenhouse gases—offering a compelling route for emissions mitigation while generating useful chemical feedstocks [58–61].

Despite its potential, widespread commercial use of DRM is hindered by the lack of stable and effective catalysts. Economically attractive carbon-based catalysts offer advantages like abundance, low cost, easy regeneration, high temperature tolerance and sulfur resistance compared to metal catalysts [62–64]. While activated carbon is commonly used for catalytic methane decomposition, biochar is gaining interest for its lower carbon footprint in hydrogen production. Several authors [65–75] have studied the synthesis, characterization and application of biochar materials as catalysts in dry reforming of methane reaction, as summarized in Table 2.

Table 2. Application of biochar materials as catalysts in dry reforming of methane reaction. (N.R. means Not Reported).

Entry	Catalyst Source	Reaction Performance	Observations ^a	Ref.
1	MW-assisted biochar prepared from cotton stalk BC-10 and BC-10 (with Fe)	CO/H ₂ = 1.02 Syngas Production = 52.5 mL/min Syngas yield = 93.2%	T _{conversion} = 800 °C MW power = max 3 KW	[67]
2	Biochar prepared from hawthorn seed BC-11	CO/H ₂ = 0.7 X _{CO2} = 68%; X _{CH4} = 84% Syngas yield = N.R.	T _{conversion} = N.R. (MW) MW power = 900 W	[68]
3	Biochar prepared from lotus stems BC-12	CO/H ₂ = 0.33 Syngas yield = N.R.	T _{conversion} = 500 °C	[69]
4	W-modified biochar, prepared from pine wood W@BC-13	CO/H ₂ = 1.08 X _{CO2} = 95%; X _{CH4} = 83% Syngas yield = N.R.	T _{conversion} = 850 °C	[70]

Table 2. Cont.

Entry	Catalyst Source	Reaction Performance	Observations ^a	Ref.
5	Co-modified biochar, prepared from oil palm shell Co@BC-14	CO/H ₂ = 0.5 X _{CO2} = 17.5%; X _{CH4} = 15% Syngas yield = 60%	T _{conversion} = 750 °C	[71]
6	Co-modified biochar, prepared from char residues Co@BC-15	CO/H ₂ = 0.45 X _{CO2} = 95%; X _{CH4} = 94% Syngas yield = 97% (For Ni@BC-16-C)	T _{conversion} = 850 °C	[72]
7	Ni-modified biochar, prepared from palm kernel shell Ni@BC-16-P and Ni@BC-16-C	CO/H ₂ = 4.6 X _{CO2} = 31%; X _{CH4} = 28% Syngas yield = 17%	T _{conversion} = 800 °C	[73]
8	Ni-Co-modified biochar, prepared from coconut shell Ni-Co@BC-17	CO/H ₂ = N.R. X _{CO2} = 94%; X _{CH4} = 97.5% Syngas yield = N.R.	T _{conversion} = 900 °C	[74]
9	Ni-Fe-modified biochar, prepared from rice husks Ni-Fe@BC-18; Ni@BC-18; Fe@BC-18	CO/H ₂ = 0.55 Syngas yield = 2.1 L/g cat	T _{conversion} = 850 °C	[75]

^a T_{conversion} represents the optimal temperature at which the highest conversion occurred.

Li prepared biochar **BC-10**, from cotton stalk [67]. The raw material was first subjected to MW irradiation (max 3 KW power) and pyrolyzed at ~800 °C for 0.5 h. The addition of metal components (K, Fe and Ni) on biochar was then carried out by mechanical mixing to produce the corresponding oxides. The experimental setup for methane reforming was mainly constituted by multi-parameter testing unit, reformer, gas feeding unit, tail gas treatment device and gas analyzer (Figure 2).

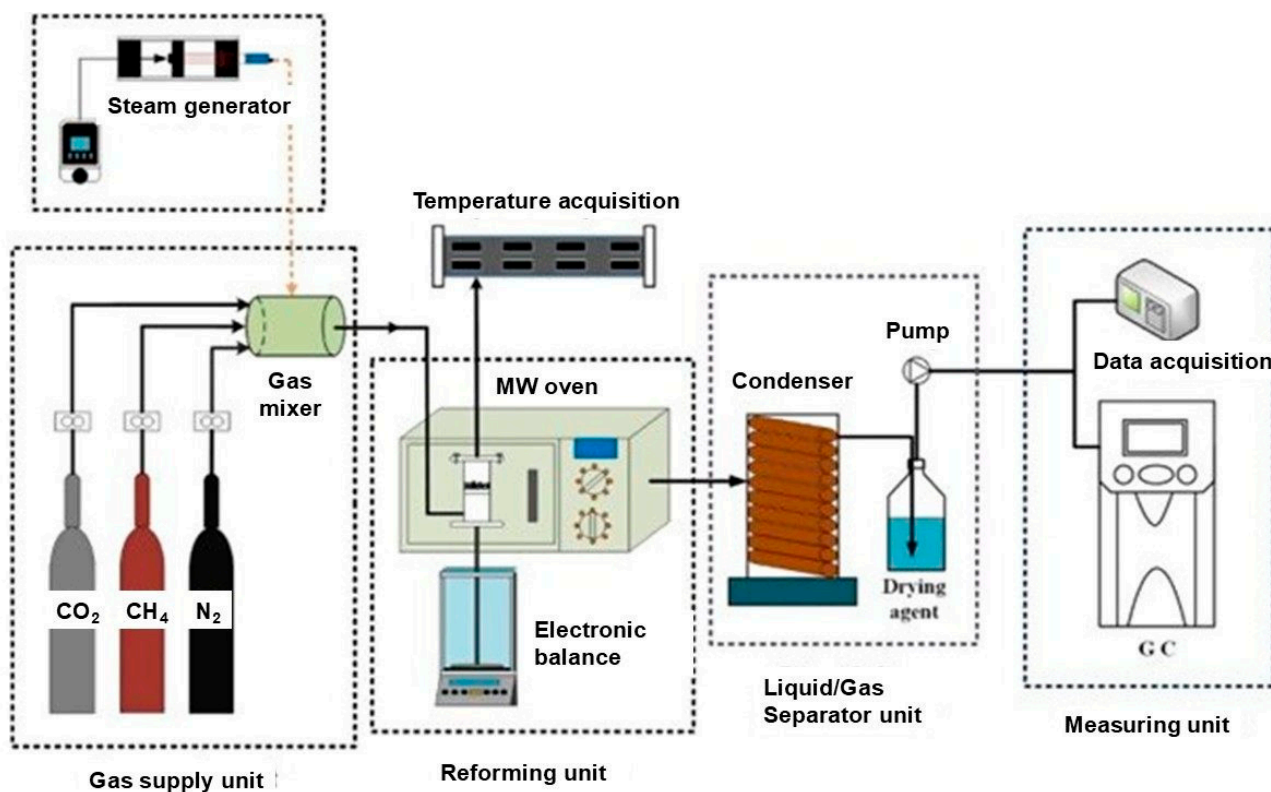


Figure 2. Schematic diagram of microwave-assisted reforming experiment setup described by Li and coworkers. Adapted from reference [67] with permission from Elsevier, Copyright 2019.

The duration of the reforming process was 2 h, and the produced gases were collected with a regular interval of 10 min [67]. The investigated temperatures for the reforming process were 700 °C, 800 °C and 900 °C under either MW heating or electric heating (HE). Methane conversion was continuously higher when MW was used as a heating source, with a more stable consumption for the whole operation time (92% with MW vs. 67% with HE, after 2 h). The best $T_{\text{conversion}}$ was 800 °C, with a 92% methane conversion, with an inlet gas molar ratio of $\text{CH}_4/\text{CO}_2 = 1:1$. The addition of Fe into biochar was proven to be the most effective to this process, causing a growth of methane conversion. Under the optimal conditions, the yield of syngas production was up to 93.2% [68].

Song also reported the microwave assisted synthesis of biochar **BC-11**, prepared from hawthorn seed [68]. The preparation method included a torrefaction pretreatment of hawthorn seed at 250, 275 and 300 °C for 1 h, followed by pyrolysis at 600 °C for 0.5 h (Table 2, entry 1). The **BC-11** biochar obtained was utilized as a catalyst in DRM under microwave irradiation at 900 W. The authors aimed to explore the influence of torrefaction and the subsequent pyrolysis process on the catalytic performance of biochar **BC-11**. They observed that torrefaction affected the composition of biochar, thus impacting the pyrolysis process. With the increase of torrefaction temperature, from 250 to 275 and 300 °C, the biochar yields also increased, enhancing deoxygenation and dehydration of solid products, thus increasing gas and liquid products. The carbon content of biochar with potential as carbon-based catalyst could exceed 80%. The torrefaction pretreatment significantly improved the catalytic performance of biochar **BC-11**. The CO_2 and CH_4 conversion rates reached ~68% and ~84%, respectively, when the torrefaction temperature was 300 °C [68].

Xiong synthesized biochar **BC-12**, using lotus stems as biomass material [69] (Table 2, entry 3). Firstly, phosphoric acid solution was used to impregnate lotus stems. After that, the mixture was dried at 105 °C for 48 h. The activation process was conducted at 500 °C under N_2 atmosphere for 90 min. The obtained catalyst was used in the co-pyrolysis of raw bamboo powder with low-density polyethylene (LDPE), which happened in a two-stage vertical furnace, in which the upper stage was the pyrolysis zone, while the other stage was the catalytic zone. Catalytic co-pyrolysis bamboo and LDPE with different LDPE ratios (0%, 10%, 20%, 30%, 40%, 50 wt%) were carried out at diverse temperatures (450, 500, 550 and 600 °C) and with a catalyst-to-feedstock ratio of 2:1. With the increase in the LDPE ratio, the solid, liquid and coke yields all showed a decreasing trend, while the gas yield was increased. At the optimum temperature of 500 °C and a LDPE ratio of 50 wt%, the volume ratio of H_2 to CO was close to 3:1, which was the gas atmosphere suitable for methanation production [69].

Yu reported the synthesis of biochar **BC-13**, using pine wood [70]. The biochar was prepared by fast pyrolysis of pine wood at 500 °C. Then, after acid washing, it was added to an ammonium tungstate aqueous solution, stirred at 80 °C for 30 min, and then oven-dried at 110 °C for 24 h. The W-enriched biochar was then placed in a furnace and subjected to carbothermal reduction at 1100 °C. The authors studied the effect of the reaction temperature on the catalytic activity of **BC-13** and product yields in CH_4/CO_2 reforming, as shown in Figure 3 [70].

The lower the reaction temperature, the lower the CH_4 and CO_2 conversions, as well as the lower the CO yield, since dry reforming is an endothermic reaction. Low feed conversion (8.9% for CH_4 and 20% for CO_2) at a low temperature (600 °C) was observed. However, the conversion of CO_2 , as shown in Figure 3a, was significantly higher than that of CH_4 , and the H_2/CO ratio varied from 0.35 at 600 °C to 0.95 at 900 °C (Figure 3b). The catalyst was found to be very stable at 850 °C for a period of over 500 h. The CH_4 and CO_2 conversions increased steadily during the first 20 h at 850 °C and then stabilized at

95% and 83%, respectively, with a CO yield of 91% and a CO/H₂ ratio after 500 h run-time remaining at around 1.10–1.15 [70].

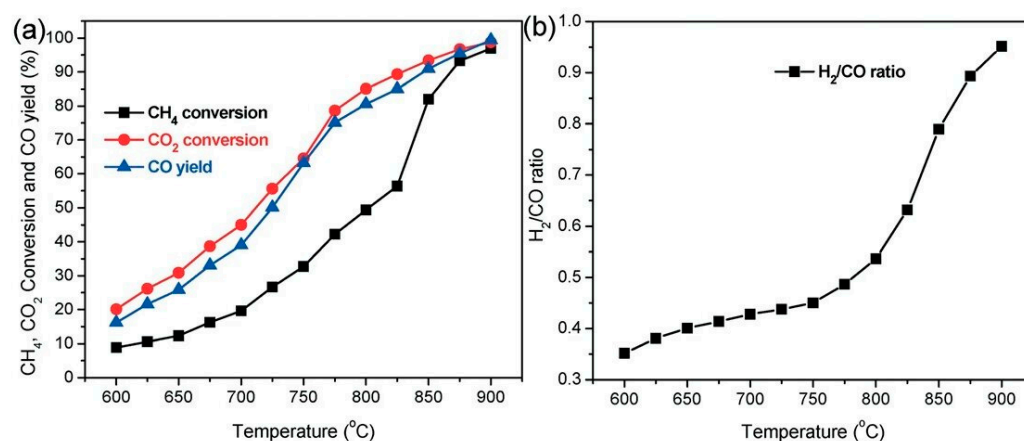


Figure 3. Effect of reaction temperature on feed conversion and CO yield during CH₄/CO₂ reforming over tungsten carbide nanoparticles in a biochar matrix at a CH₄/CO₂ ratio of 1, and 0.5 MPa: (a) feed conversion and (b) H₂/CO ratio. Adapted from reference [70] with permission from the Royal Society of Chemistry, Copyright 2015.

Amin [71] immobilized cobalt catalyst precursors onto commercial oil palm shell activated biochar via a wet impregnation method by varying the Co ratio of 6.0 and 14.0 wt% to produce **Co@BC-14**. The dry reforming of methane was performed using a micro reactor system under the condition of 10,000 mL/h.g-cat, 3 MPa, CH₄/CO₂ feeding ratio of 1.2:1.0 and temperature range from 650 to 750 °C. The catalyst **Co@BC-14** was compared to the commercial zeolite-type catalyst ZSM-5, and both catalysts exhibited lower conversions of CO₂ and CH₄ at low temperatures (Table 2, entry 5). With the concomitant increase in temperature from 650 to 750 °C, and cobalt loadings, the **Co@BC-14** catalyst having 14 wt% Co exhibited higher conversions of CO₂ and CH₄ ($X_{\text{CO}_2} = 17.5\%$; $X_{\text{CH}_4} = 15\%$), vs. ZSM-5 catalyst ($X_{\text{CO}_2} = 7.0\%$; $X_{\text{CH}_4} = 7.5\%$). **Co@BC-14** also gave higher yields of H₂ and CO (60% combined) and a 1:2 CO/H₂ ratio, after 1.5 h. The authors attributed these results to a higher surface area of **Co@BC-14** than Co-ZSM-5, which subsequently rendered higher activity for the reforming of methane.

Benedetti reported the utilization of char residues collected from small-scale biomass transformation plants to produce the biochar **Co@BC-15** (Table 2, entry 6) [72]. The char-supported cobalt catalysts were synthesized using the wetness impregnation method, with cobalt loadings of 10, 15 and 20 wt.% and calcined in a muffle furnace at 400 °C for 4 h. Authors observed that the **Co@BC-15** catalyst having 10 wt% Co was the optimal Co loading. Hence, two additional catalysts were prepared: one treated with HNO₃ and another with MgO. This improved CO₂ and CH₄ conversions, and HNO₃ treatment resulted in slightly higher activities and yields. However, the presence of MgO drastically increased the conversions and the yields up to 95 and 94% for CO₂ and CH₄ conversions, and 44 and 53% for H₂ and CO yields. This was attributed to the capacity of MgO in hindering carbon deposition, the main reason for DRM catalyst deactivation. Furthermore, the catalyst remained stable throughout the experiments [72].

Lam synthesized the biochar **Ni@BC-16** from palm kernel shell [73], which was first subjected to carbonization in a MW oven (700 W) for 25 min followed by chemical (KOH and NaOH) or physical (H₂O) activation (Table 2, entry 7). Both types of activation were performed by carbonization in a MW oven (700 W) for 25 min. The resulting biochar materials were then Ni-modified by the impregnation method with calcination at 300 °C to produce **Ni@BC-16-C** (chemically activated biochar) and **Ni@BC-16-P** (physically activated

biochar). The catalysts were tested in the methane dry reforming reaction to evaluate their performance, and it was found that the conversion efficiency of methane (CH_4) was decreased when the process time was increased from 0.25 to 2 h for both catalysts, indicating that the conversion of methane should be performed in shorter process times to obtain higher conversion efficiency [73]. The authors attributed the lower conversion efficiency of methane recorded at longer process times to coke formation on the catalyst surface, which would decrease methane conversion. Regarding CO_2 , **Ni@BC-16-C** showed an increasing trend of CO_2 conversion rate (from 18 to 31%) as the process time was increased, but **Ni@BC-16-P** recorded an opposite trend (from 31 to 26%).

Li reported the catalyst **Ni-Co@BC-17**, a bimetallic modified biochar prepared from coconut shell through the solid-phase method [74]. The commercially available coconut shell-based activated carbon and polyvinylpyrrolidone (PVP) were mixed in diverse ratios and pre-treated. Thereafter, metal salts of $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ were mechanically blended with the previously treated mixtures and underwent a complete grinding for developing the final catalysts by a solid-phase method, implemented at 600 °C, under nitrogen atmosphere, for 1 h. Catalysts with Ni/Co = 2:1, 1:1 and 1:2 were obtained (Table 2, entry 8). Bimetallic catalysts with Ni/Co ratios of 1:2 and 1:1 were the best-performing biochars showing similar catalytic selectivity towards CH_4 and CO_2 conversions, with average conversions of 94.0% and 97.5% respectively, at 900 °C for 12 h.

Shen used rice husks as biomass to develop **Ni-Fe@BC-18** catalysts [75]. The biochar was prepared by slow pyrolysis at a temperature of 700 °C in a N_2 atmosphere. The wet impregnation method was used to immobilize diverse Ni and/or Fe loads. After impregnation and drying overnight at 105 °C, the char-supported catalysts were calcined in air at 600 °C for 1 h (Table 2, entry 9). The authors observed that monometallic **Ni@BC-18** exhibited much higher reforming activity of hydrocarbons than monometallic **Fe@BC-18** catalysts. The authors also observed a detrimental effect with the presence of Fe. While **Ni-Fe@BC-18** outperformed monometallic Fe catalysts, its activity was lower than that of the monometallic Ni catalyst. The authors attributed this reduced performance to the co-catalytic function of Fe. Since Fe has a higher oxygen affinity than Ni, the addition of Fe to Ni catalysts can increase the coverage of oxygen atoms during the reforming reactions, preventing tar conversion. The catalytic activity of Ni-Fe catalysts for tar elimination could be concluded in the following order: **Ni-BC@18** > **Ni-Fe@BC-18** > **Fe@BC-18** > **BC-18**, with the best catalyst reaching a CO/H_2 ratio = 0.55 and a *syngas* yield = 1.7 L/g cat [75].

Among the catalysts studied, **Co@BC-15** (Entry 6, Table 2) exhibits the highest *syngas* yield (97%) along with excellent CO_2 and CH_4 conversions (95% and 94%) at 850 °C, making it a standout in terms of overall efficiency. However, **Ni@BC-16-C** (Entry 7, Table 2) demonstrates the highest CO/H_2 ratio (4.6), which may be desirable for specific downstream applications, while **BC-10** and **W@BC-13** showed the closest-to-unity CO/H_2 ratio, which is favorable, for instance, for hydroformylation reactions. The data also indicate that transition metal modification of biochar, especially with Co and Ni, significantly enhances reforming performance, particularly at higher temperatures.

4. Catalytic Biomass Upgrading Reactions

CO_2 can be applied as a green reagent in catalytic biomass upgrading, enabling its incorporation into biomass-derived intermediates to produce value-added chemicals such as carbonates, carboxylic acids, *syngas* and fuels. Heterogeneous catalytic systems based on carbon materials, particularly biochar, have shown high potential for these transformations. Their catalytic behavior is strongly influenced by the biochar's feedstock, ash content, and surface composition, especially the presence of alkali metals and oxygen-containing

functional groups. These characteristics allow both CO₂ activation and product selectivity, making biochar a versatile platform for CO₂ valorization in biomass-derived systems.

Zhao [76] investigated the catalytic behavior of rice husk-derived biochar (**BC-19**) in biomass tar upgrading through reforming with CO₂ and H₂O. The focus was on evaluating the effects of CO₂ and H₂O, both as reforming agents and as modulators of the biochar structure during the catalytic process, to elucidate the transformation mechanisms of tar and the role of biochar under reactive gas atmospheres. The study utilized a two-stage reactor system, consisting of a fluidized bed for tar generation and a fixed bed for reforming, where biochar derived from biomass pyrolysis served as the heterogeneous catalyst (Figure 4).

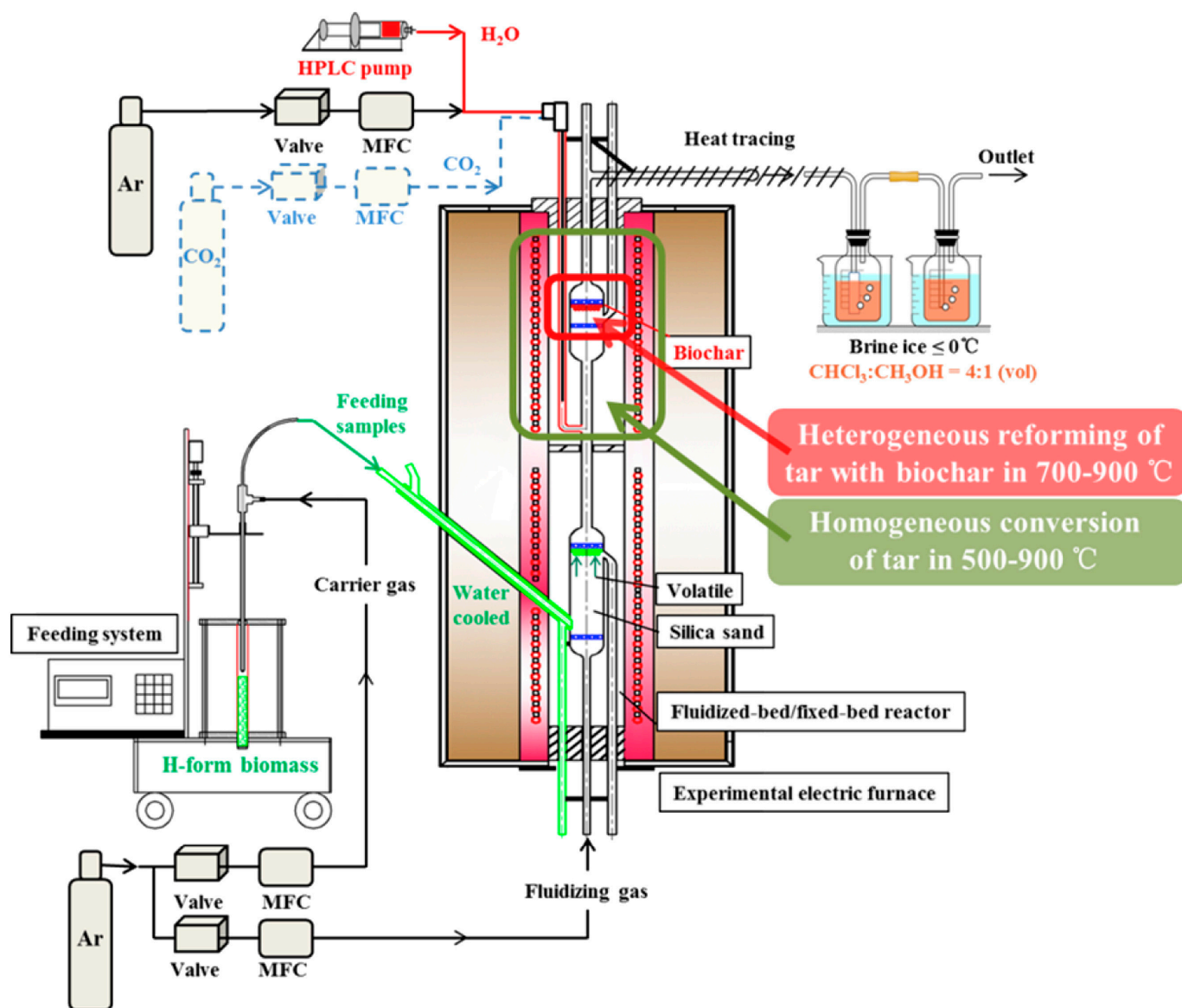


Figure 4. Schematic diagram of a two-stage fluidized bed/fixed bed reactor for the homogeneous conversion and heterogeneous reforming of biomass tar, described by Zhao and collaborators. Adapted from reference [76] with permission from Elsevier, Copyright 2017.

The biochar **BC-19** synthesis involved conventional pyrolysis, resulting in a porous carbonaceous material rich in active sites capable of interacting with tar compounds. The catalytic experiments were performed at high temperatures ranging from 700 to 900 °C to ensure significant tar conversion, particularly of polycyclic aromatic hydrocarbons (PAHs). It was observed that both CO₂ and H₂O atmospheres significantly contributed

to the homogeneous and heterogeneous reforming of tar. In homogeneous reactions, 15 vol.% H₂O exhibited a more pronounced effect on tar conversion than 29 vol.% CO₂, indicating a stronger reforming capability, especially in degrading aromatic compounds. In heterogeneous reforming reactions carried out at 800 °C with the biochar catalyst, the tar yield decreased substantially, demonstrating high catalytic activity. Under an inert Ar atmosphere, **BC-19** alone reduced the tar yield to 3.63%. The introduction of 5 to 15 vol.% H₂O further decreased the tar yield progressively from 2.60% to 1.38%, while increasing CO₂ concentration from 10 to 29 vol.% led to a decrease from 3.02% to 1.89% (Figure 5A) [76]. These findings underscore the synergistic role of CO₂ and H₂O in enhancing the reforming efficiency of biochar. Additionally, both CO₂ and H₂O were found to induce structural changes in the biochar catalyst, contributing indirectly to improved reforming performance. These changes included the formation of additional oxygen-containing surface functional groups and the transformation of smaller aromatic ring structures within the biochar matrix into larger, more graphitized domains (Figure 5B). This structural evolution enhanced the interaction between the tar molecules and the biochar surface, particularly facilitating the reforming of aromatic tar components, such as compounds with five-carbon rings or larger polyaromatic systems. The reactivity was notably higher for aromatic tar species compared to non-aromatic ones, indicating selectivity towards the breakdown of complex aromatic structures under CO₂/H₂O atmospheres [76]. This study demonstrated that biochar serves as an effective catalyst for the upgrading of biomass tar via CO₂ and H₂O reforming. Its performance is significantly influenced by the operating temperature, gas composition, and the resulting physicochemical modifications to the biochar structure. The high tar conversion rates, particularly under increasing concentrations of CO₂ and H₂O, and the selectivity towards aromatic components highlight the potential of biochar-based catalysts in biomass valorization processes involving CO₂ utilization [76].

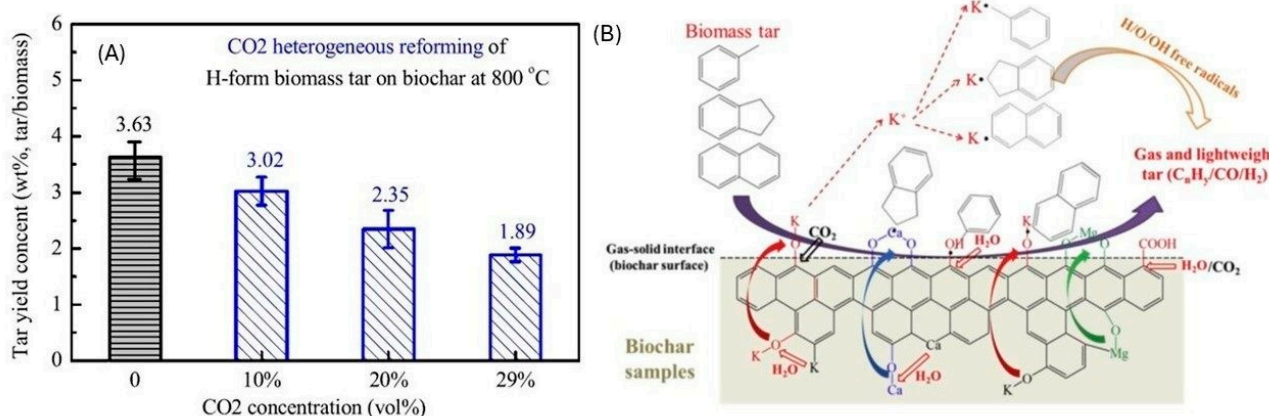
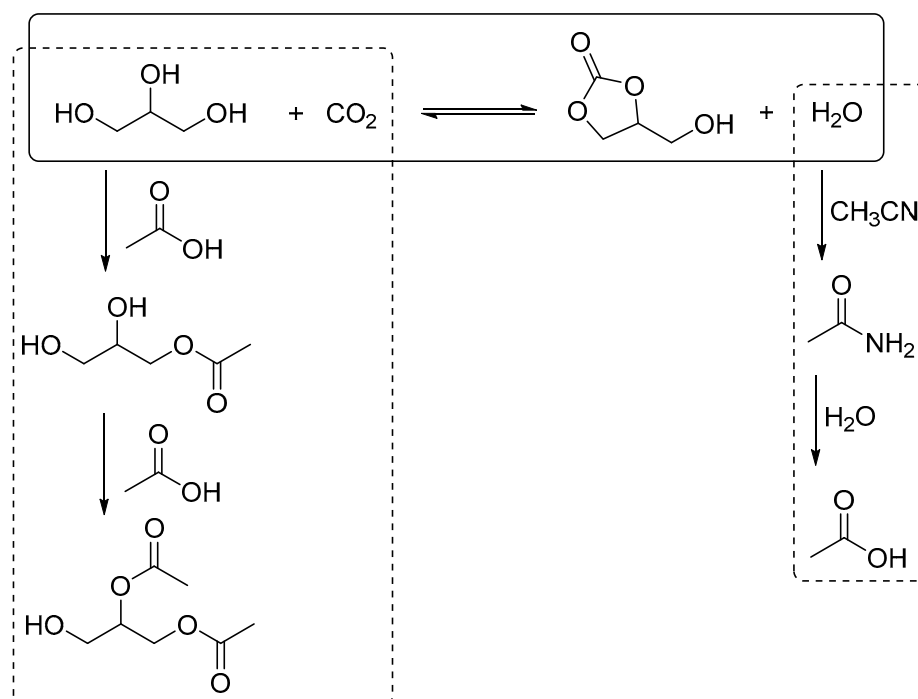


Figure 5. (A) Tar yield during CO₂ heterogeneous reforming on biochar **BC-19**; (B) Proposed mechanism of heterogeneous reforming of biomass tar over biochar **BC-19** in the presence of H₂O and CO₂ at 800 °C. Adapted from reference [76] with permission from Elsevier, Copyright 2017.

McGregor [77] explored the catalytic potential of biochar materials derived from different biomass sources for the carboxylation of glycerol with CO₂, using acetonitrile as a dehydrating agent. This reaction led to the formation of glycerol carbonate and acetins, namely diacetin and triacetin, products of interest in green chemistry and CO₂ valorization, Scheme 11. This work focused on correlating the physicochemical properties of biochars with their catalytic activity, particularly considering the influence of ash and carbon content, and the composition of key mineral elements such as potassium. Biochars were synthesized from three main biomass feedstocks, including soft wood (**BC-20-SWP**), oilseed rape (**BC-20-OSR**) and rice husk (**BC-20-RH**), through pyrolysis at different temperatures. The resulting

catalysts varied in surface area, ash content, carbon structure and elemental composition. High-ash biochars, particularly those derived from OSR and RH, exhibited notable catalytic performance, while low-ash materials, including soft wood biochar and commercial activated carbon, were largely inactive for glycerol carbonate and triacetin production [77]. For example, **BC-20-OSR 550** (prepared at 550 °C) with 17.1 at.% surface potassium yielded 0.0340 mol/L of glycerol carbonate and 7.16×10^{-4} mol/L of triacetin, while **BC-20-RH 550** achieved 0.0482 mol/L glycerol carbonate and 4.79×10^{-4} mol/L triacetin. In contrast, **BC-20-SWP 550** produced negligible quantities of these products, underscoring the importance of mineral content, particularly potassium, in catalytic activity. Demineralization of biochar with hydrochloric acid confirmed this trend: the removal of potassium reduced glycerol carbonate yields by over 94% and entirely suppressed triacetin formation, highlighting potassium's essential role in catalysis. Additionally, isolated biochar ash samples, despite low surface areas (19–38 m²/g), retained catalytic activity (e.g., **BC-20-OSR 700-ash** yielded 0.145 mol/L glycerol carbonate and 4.38×10^{-3} mol/L triacetin), further confirming the contribution of ash minerals [77]. Interestingly, while ash-rich samples were dominant in carbonate and triacetin production, biochars with low ash but graphitic carbon content were effective in the formation of diacetin. Soft wood biochar (**BC-20-SWP 550**), with a more graphitic structure ($AD_1/AG = 1.32$), yielded 1.25×10^{-3} mol/L of diacetin, outperforming activated carbon despite having a lower surface area, suggesting graphitic carbon may play a role in diacetin selectivity. The findings from this study provide strong evidence that biochar catalytic performance in CO₂ utilization reactions is highly dependent on feedstock origin, ash content, and specific mineral elements such as potassium. The demonstrated ability of biochar ash and carbon matrix to selectively catalyze the formation of various value-added products reinforces the potential of tailored biochar materials in sustainable CO₂ conversion processes [77].



Scheme 11. Synthesis of glycerol carbonate from glycerol and carbon dioxide, and reaction pathways for the formation of mono- and diacetins, in the presence of acetonitrile as a dehydrating agent.

5. Conclusions and Perspectives

Biochar has emerged as a highly promising material for catalytic CO₂ valorization, offering a sustainable and efficient alternative to conventional catalytic systems. Its intrinsic characteristics—renewability, high surface area, tunable porosity and adaptable surface chemistry—make it particularly suitable for heterogeneous catalysis in various CO₂ conversion processes. Applications of biochar-based catalysts include several key CO₂ utilization strategies, including cycloaddition to epoxides to produce cyclic carbonates, dry reforming of methane to generate syngas, catalytic biomass upgrading to obtain high-value chemicals and fuels, and CO₂ reduction reactions to form carbon monoxide, methane, methanol or formate, all contributing to the development of low-carbon energy systems. A critical factor influencing biochar's catalytic performance is its source. The nature of the biomass feedstock and the pyrolysis conditions directly affect the chemical composition, surface functionality, and ash content of the resulting biochar. These attributes, in turn, determine catalytic activity, selectivity and stability. Feedstocks rich in inorganic content can enhance CO₂ activation through the presence of alkali and alkaline earth metals, while biochar derived from lignocellulosic biomass may benefit from a high degree of graphitization or oxygen-containing groups. Understanding and controlling these source-dependent characteristics is essential for the rational design of high-performance catalysts. Compared to traditional homogeneous catalysts, biochar-based systems offer key advantages such as enhanced recyclability, cost-effectiveness, reduced toxicity and environmental compatibility. Their ability to anchor active sites and resist metal leaching further improves operational stability and reuse. Moreover, targeted modifications—such as heteroatom doping or metal deposition—can fine-tune the surface chemistry and acid–base properties, thereby increasing catalytic efficiency and product selectivity. However, challenges remain regarding property variability, metal leaching in doped systems and the scalability of production processes. Addressing these limitations will require the development of standardized protocols for biochar synthesis and characterization, as well as deeper insight into structure–performance relationships.

The modulation of biochar structure, through the incorporation of heteroatoms (N, O) and/or metals like (Ni, Mg, Co) or acidic or basic functionalization, significantly influenced its catalytic activity in CO₂ addition reaction to epoxides, demonstrating the correlation between structure and reactivity. Nitrogen-doped biochars exhibited high selectivity for CO₂ conversion, attributed to the presence of pyridonic and quaternary groups, which act as basic sites for CO₂ activation. Oxygen-enriched porous materials demonstrated epoxide activation capability via hydrogen bonding, highlighting the crucial role of acidic functions in reactivity. Additionally, the incorporation of MgO into hierarchical biochars resulted in a favorable balance between acidity and basicity, providing superior catalytic efficiency. Another determining factor in catalytic activity was porosity and CO₂/N₂ adsorption selectivity.

Regarding dry reforming of methane, bimetallic catalysts, particularly those with Ni/Co, demonstrated superior performance among the biochar-supported systems studied, which exhibited comparable and balanced selectivity toward both CH₄ and CO₂ conversions. The presence of cobalt may enhance the dispersion and stability of nickel particles, reduce carbon deposition, and improve the overall redox behavior of the catalyst—factors that may contribute to increased activity and long-term durability under DRM conditions.

Advanced functionalization techniques will be crucial to tailoring catalytic properties for specific applications, while comprehensive life cycle assessments are needed to validate the environmental and economic benefits of these materials. Biochar represents a versatile and renewable platform for catalytic CO₂ valorization. Emphasizing the role of biomass source selection and precise structural modifications will be pivotal for optimizing catalytic

performance and ensuring the reproducibility and scalability of biochar-based systems in sustainable chemical processes. Moreover, the inherent recyclability and robust structural stability of biochar-based materials make them particularly attractive as sustainable catalysts, further strengthening their role in the efficient and environmentally responsible conversion of CO₂ into high-value products.

Despite the promising catalytic application of biochar-based materials, there are still two main challenges that require scientific attention, namely, catalyst deactivation and scalability.

Catalyst deactivation still remains a prominent concern for the economic sustainability of CO₂ valorization reactions over biochar-supported catalysts due to the following: (i) Coking, or carbon deposition, is commonly described where carbonaceous compounds build upon active sites and mechanically obstruct them, particularly in the case of CO₂ reduction reactions like dry reforming of methane; (ii) Sintering, which describes the agglomeration and growth of metallic nanoparticles, e.g., Ni, Co, at elevated temperatures, greatly diminishes the catalyst's active surface area; (iii) Metal leaching, a phenomenon that may take place in the liquid phase or certain gaseous phase conditions, leads to the detachment of active metal parts from the biochar support; (iv) Poisoning occurs due to the occlusion of active sites by detrimental species like sulfur compounds or halides which cannot be removed; and (v) Fouling, which refers to the physical blocking of pores by non-carbonaceous byproducts and structural degradation of the biochar support itself, can lead to reduced surface area and altered properties.

Moreover, the challenge of batch-to-batch reproducibility makes scaling biochar production extremely difficult. The multitude of biomass feedstocks available is a plethora of benchmarks of varying chemical structures including lignin, water content and ash residue, all of which will create different biochars under identical pyrolysis conditions. The varying characteristics of biochars makes quality and standard set prediction, which is critical for market acceptance, considerably complex. Such variations in reproducibility might prove to be a constraint for industrial-scale catalytic applications.

In sum, addressing these challenges requires a systematic approach to synthesis and characterization, including (i) standardizing feedstock and (ii) implementing advanced pyrolysis technologies with real-time monitoring for consistency. These parameters are nowadays defined by the International Biochar Initiative (IBI—<https://biochar-international.org/>, URL accessed on 5 June 2025) and the European Biochar Certificate (EBC—<https://www.european-biochar.org/en>, URL accessed on 5 June 2025), which are developing international standards to establish consistent terminology, analytical methods and quality benchmarks, creating a framework to classify biochar by its properties and intended use. This will bridge the gap between diverse raw materials and consistent product quality, ultimately paving the way for the true industrial scaling of biochar production.

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